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Title: Vision-Guided Automated Foosball Opponent

Motivation: Many games have an accessible method of practice against a non-human opponent; however, foosball is a game that requires at least two players to enjoy. The ECE common room currently houses a foosball table, which students use as a break activity when studying to help refresh and de-stress. This project aims to create an automated foosball opponent that allows for solo play and independent practice while also serving as an interactive platform for students to explore concepts in computer vision, mechatronics, and real-time control.

Goal: Design and implement a foosball table extension that automates one player's rods, allowing for primarily defensive but potentially offensive play based on the ball's trajectory.

Brief Description: Computer vision will be used to track the position and trajectory of the ball, along with the locations of the "foosmen" (figurines on the rods of a foosball table that are used to strike the ball). Additionally, it will determine the optimal rod positions required for the automated player to intercept and play the ball. These rods will be motorized with a mechatronic setup to enable both rotational and translational movements controlled through software. This automated opponent will be designed to defend in the opposite direction, with future extensions exploring strategic plays based on the positioning of the human opponents' "foosmen".

Anticipated Outcome: A functional foosball table that automates one player's movement on at least two defensive lines, reliably matching an amateur's ability for casual play. The intention is for this automated foosball table to be placed in the ECE common room, providing all students and staff with the opportunity to enjoy with or without a friend.